An Interpretable, Calibrated, and Auditable Behavioural Risk Model for Online Gambling Harm Detection

with privacy-preserving operator deployment and in-browser recalibration

Kahlil Simeon-Rose
Carson College of Business, Washington State University, Everett, WA, USA

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Abstract

We describe the risk model and system architecture of an open, white-label responsible-gambling (RG) analytics platform that scores online gambling accounts for behavioural risk. The model is interpretable by construction: a player's 0–100 risk index is a non-negative weighted mean of thirteen literature-anchored behavioural markers, and a separate discrete-time hazard layer maps that index to a calibrated probability of the harm proxy within twelve months. Marker weights are fitted on a large longitudinal research cohort of online bettors (subsequent RG-coded account closure as the outcome) under a leakage-aware prospective design, blending a monotonic (non-negative) multivariate logistic fit with per-marker univariate discrimination so that redundancy is down-weighted without eliminating any clinically grounded marker. The deployed rule attains ROC-AUC 0.673 against ever-exclusion and 0.711 at the twelve-month horizon (multivariate ceiling 0.688; PR-AUC 0.109 against a 0.048 base rate), with well-calibrated twelve-month probabilities (Brier 0.023, ECE 0.003). A distinct human-in-the-loop overlay folds structured staff observations into the score with incremental-information weighting, a bounded corroboration uplift, and a standing disclosure floor that clears only on human follow-up. The platform is designed for operator self-deployment: data onboarding with automatic column mapping, entirely client-side re-fitting and recalibration on operator-labelled outcomes (no data egress), a tamper-evident hash-chained audit trail, model cards with composition versioning, and fairness audits that include the discretionary observation channel. All components ship as an open reference implementation on synthetic data.

1 Introduction

Gambling operators are increasingly required—by regulation, licensing condition, or their own safer-gambling commitments—to identify accounts at elevated risk of gambling-related harm early enough to intervene. Vendor solutions in this space are typically proprietary: model structure, training labels, validation results, and failure modes are not inspectable by the operator, the regulator, or the research community. This sits uncomfortably with two facts. First, high-stakes decisions about individual people are involved, which argues for models that are interpretable rather than post-hoc explained [18]. Second, the underlying science—behavioural markers of harm in account data—originates in the *open* literature [1–3, 5], so there is no fundamental reason the operational layer must be opaque.

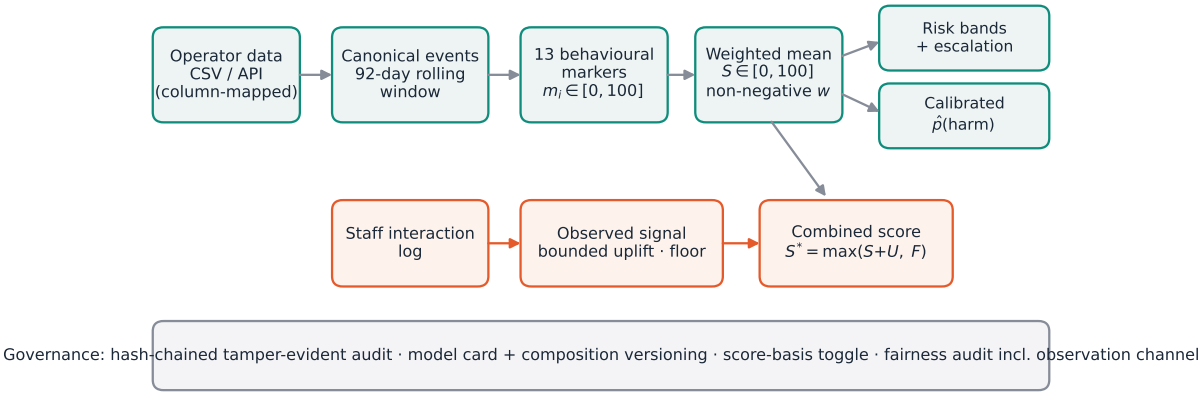


Figure 1: System architecture. The data lane (top) computes thirteen behavioural markers over a rolling 92-day window and aggregates them into the ordinal index S, which maps to monitoring tiers and, through a separate calibration layer, to $\hat{p}$(harm). The human lane (middle) folds structured staff observations into a combined score via a bounded uplift and a disclosure floor. All score-affecting actions feed the governance layer (bottom).

This work is an attempt to build the operational layer in the open. This paper documents its risk model and the design decisions that make the system deployable by an RG team rather than a machine-learning team. Our contributions:

1. **An interpretable, calibrated two-layer score** (§5): an ordinal 0–100 index (a non-negative weighted mean of thirteen behavioural markers) that drives five monitoring tiers, plus a separately reported calibrated probability of the harm proxy. Keeping the layers distinct avoids the tier collapse that occurs when rare-outcome probabilities are used directly as scores.
2. **A defensible weight-fitting recipe for collinear marker sets** (§5.2): blending a monotonicity-constrained multivariate fit with univariate discrimination, so weights are redundancy-aware yet no canonical marker is zeroed by collinearity—preserving face validity for regulators and practitioners.
3. **A human-in-the-loop overlay** (§6) that treats the staff interaction log as a model input: observed markers corroborate the data-derived score under *incremental-information* weighting (confirming what the data already shows adds little; data-blind emotional/social signal counts fully), and critical disclosures impose a standing floor cleared only by human follow-up.
4. **Governance as a first-class subsystem** (§8): a tamper-evident hash-chained audit of every score-affecting action, model cards with composition versioning [19], a global toggle between the validated data-only score and the combined score, and fairness audits that cover the discretionary observation channel.
5. **Privacy-preserving operator deployment** (§7): column-mapped data onboarding and full model re-fitting/recalibration executed in the operator's browser, so account-level data never leaves their environment.

2 Background

Behavioural markers. Longitudinal studies of online-gambling account data established that early betting behaviour separates players who later exhibit harm-adjacent outcomes: Braverman and Shaffer [2] identified frequency, intensity, variability, and trajectory markers in the first month of play; later work added cross-game involvement ("breadth") as an independent signal [3, 4]. Payment behaviour is similarly informative: Haeusler [5] showed deposit and withdrawal patterns (including reversed withdrawals and the number of payment methods) predict future self-exclusion, and account holders flagged by corporate RG programmes show distinctive transaction patterns [6]. Machine-learning replications reach comparable conclusions with deposit-frequency features among the strongest predictors [7–10].

Labels. There is no direct ground-truth label for gambling harm in account data. The pragmatic proxy is voluntary self-exclusion or RG-coded account closure, but it is *under-ascertained*: many harmed players never self-exclude, and motivations for closure vary [11]. Formally this is a positive–unlabeled (PU) learning problem [14]: observed positives are reliable, but negatives are a mixture. This shapes evaluation (rank metrics over accuracy; recall-oriented operating points; false positives are expected rather than errors) and communication (the model detects *similarity to players who later self-excluded*, not harm itself).

Interpretability and calibration. For consequential decisions, Rudin [18] argues for models interpretable by construction rather than post-hoc explanation of black boxes. Independently, scores used for resource allocation should be *calibrated*—predicted probabilities should match observed frequencies [15–17]. Both properties are design requirements here.

3 Data and label construction

We fit default weights on a large longitudinal account-level cohort of online bettors assembled for research use; the fitting extract is not redistributed with the product.[1] It combines daily betting aggregates, deposit and withdrawal transactions, and RG-event records, structurally comparable to the account data used throughout the behavioural-markers literature [1, 2].

Prospective windows. Each account is anchored at its first observed activity; features are computed over the following 90 days. The label is an RG-coded account closure (typically customer-initiated self-exclusion) occurring *strictly after* the feature window, with event timing retained for the hazard layer (§5.3). Accounts whose closure falls inside the window are dropped as ambiguous rather than labelled, eliminating outcome leakage into features. The runtime product mirrors this design with a 92-day rolling window.

Modelling set. Roughly ten thousand accounts, with outcome prevalence of approximately 4.8%. Demographic attributes (age, sex) are *deliberately excluded* from the deployed scorer (§8); an exploratory model including them is reported only as a reference point.

[1] The shipped product contains no cohort records: its demonstration portfolio is synthetic, generated from aggregate distributions only (§9).

Table 1: Markers, fitting proxies, univariate discrimination of later RG closure (oriented AUC and Gini), and deployed weights (w_i, normalised to mean ≈1 over fitted markers). Priors have no cohort analog.

Marker	Cohort proxy	AUC	Gini	w_i	Evidence
Increasing loss tolerance	mean daily stake	0.644	0.288	2.775	[2]
Impulsive transactions	deposits per active day	0.627	0.255	2.232	[5, 7]
Deposit-frequency escalation	deposit count	0.632	0.263	1.642	[5, 8]
Repeated loss of winnings	reversed-withdrawal amount	0.558	0.116	1.065	[5, 6]
Chasing losses	net loss over window	0.589	0.178	0.813	[2, 6]
Longer session durations	— (prior)	—	—	0.636	[13]
Late-night play	— (prior)	—	—	0.636	[13]
Inability to stop	— (prior)	—	—	0.636	[13]
Payment-method switching	distinct payment methods	0.550	0.101	0.459	[5, 10]
Deposit volatility	CV of daily stake	0.532	0.064	0.292	[2]
Play frequency	active days	0.528	0.056	0.255	[1, 2]
Shift to faster games	products played	0.515	0.029	0.250	[3, 4]
Increasing bet sizes	stake-trajectory slope	0.509	0.018	0.215	[2]

4 Behavioural markers

The engine computes thirteen markers, each a deterministic transform of the account's window mapped to a bounded $[0, 100]$ sub-score with a human-readable explanation (e.g. "returned to play after 31 of 33 significant loss days"). Table 1 lists each marker, the cohort feature used as its fitting proxy, its standalone discrimination of the label, and its fitted weight. Ten markers have direct analogs in the cohort; three (session duration, late-night play, in-session loss-of-control) have no analog in betting-and-payments data and enter as *informed priors*, grounded in practitioner codes and problem-gambling screens (e.g. PGSI loss-of-control items [13]) and assigned the median fitted weight. They are labelled as priors in the product's model card.

5 The risk model

5.1 Aggregation

Let $m_i \in [0, 100]$ be the marker sub-scores and $w_i \geq 0$ their weights. The ordinal risk index is the weighted mean

$$S \;=\; \frac{\sum_i w_i\, m_i}{\sum_i w_i} \;\in\; [0, 100]. \tag{1}$$

Monotonicity is structural: no marker can lower risk, and each marker's contribution $w_i m_i$ is directly displayable, which is what the product's explanation panels show. An earlier hand-set aggregation (a 0.62/0.38 blend of the weighted mean with the mean of the top-3 markers) was removed when weights were fitted: severity concentration is now expressed through the fitted weights themselves rather than a second heuristic.

5.2 Weight fitting

Fitting proxies are $\log(1+x)$-transformed, standardised, and oriented so that larger values indicate risk. A multivariate logistic regression is fitted under a non-negativity constraint with an ℓ_2

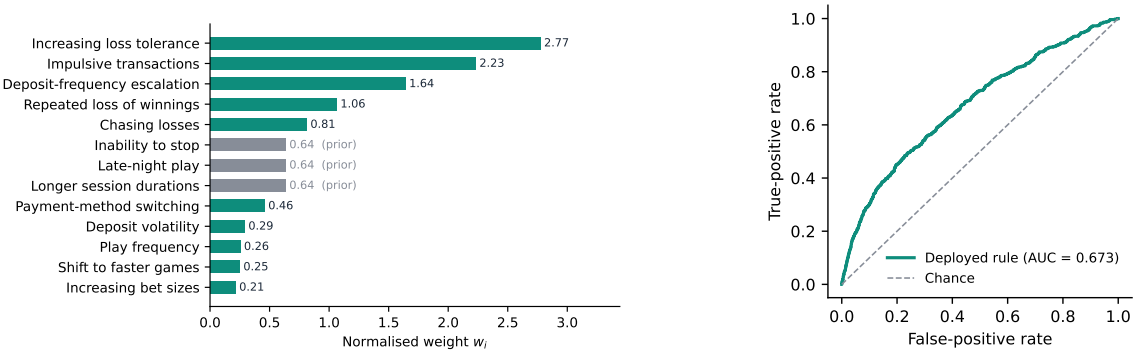


Figure 2: Left: deployed marker weights (grey = literature priors with no cohort analog). **Right:** ROC of the deployed weighted-mean rule on the fitting cohort.

penalty:

$$(b, \mathbf{w}_{\mathrm{mv}}) \;=\; \arg\min_{b,\ \mathbf{w}\geq 0} \; \tfrac{1}{n}\textstyle\sum_j \ell\big(y_j,\ \sigma(b + \mathbf{x}_j^{\top}\mathbf{w})\big) \;+\; \lambda\|\mathbf{w}\|_2^2. \tag{2}$$

Constrained multivariate fitting is redundancy-aware but, under collinearity, drives some canonical markers to exactly zero (in this cohort: chasing losses, play frequency, deposit volatility, payment switching). A scorer that assigns *zero* weight to chasing losses would be statistically defensible yet unacceptable on face validity to practitioners and regulators, and brittle if a zeroed marker's correlated proxy is missing in an operator's data. We therefore blend with per-marker univariate discrimination $g_i = 2\max(\mathrm{AUC}_i,\ 1{-}\mathrm{AUC}_i) - 1$:

$$\mathbf{w} \;=\; \mathcal{N}\Big(\tfrac{1}{2}\,\mathcal{N}(\mathbf{w}_{\mathrm{mv}}) + \tfrac{1}{2}\,\mathcal{N}(\mathbf{g})\Big), \qquad \mathcal{N}(\mathbf{v}) = \mathbf{v}\big/\,\mathrm{mean}(\mathbf{v}_+), \tag{3}$$

yielding the weights of Table 1 (Figure 2): every literature-grounded marker retains non-trivial weight, while redundant markers are substantially down-weighted relative to their univariate rank.

5.3 Calibration and score presentation

The probability layer is a *discrete-time hazard* on the index. With months $j = 1, 2, \ldots$ after the feature window grouped into pieces $\pi(j) \in \{1\text{–}3, 4\text{–}6, 7\text{–}12, 13\text{–}24, 25\text{–}36\}$,

$$h(j \mid S) \;=\; \sigma\big(\alpha_{\pi(j)} + \beta\, S/100\big), \qquad \hat{p}_H \;=\; 1 - \prod_{j=1}^{H}\big(1 - h(j \mid S)\big), \tag{4}$$

reported at the H=12-month horizon. Events are RG-coded closures dated after the window; accounts are censored at the administrative end of the records or at 36 months. This replaces the earlier binary layer and fixes its two defects: short follow-up no longer masquerades as a clean negative, and a closure three weeks after the window is no longer the same label as one three years later. The fitted monthly baseline ($\hat{\alpha} = -9.03,\ -8.72,\ -8.61,\ -9.77,\ -10.12$; $\hat{\beta} = 4.50$) peaks in months 4–12 and declines thereafter; the single slope on S is a proportional-hazards-style assumption accepted for interpretability.

Because the outcome is rare, twelve-month probabilities are small even for the riskiest accounts ($\hat{p}_{12} \approx 0.5\%$ at S=20, 1.2% at S=40, 2.9% at S=60, 6.9% at S=80). Using $100\,\hat{p}$ as "the score" would compress the whole portfolio into the lowest monitoring tier. The product therefore presents *two* numbers with distinct semantics: the ordinal index S (drives tiers, queues,

Table 2: Performance on the fitting cohort (prevalence ≈4.8%). The deployed rule's discrimination is evaluated in-sample; the unblended constrained model is additionally evaluated out-of-fold (5-fold). Calibration metrics are 5-fold out-of-fold at the twelve-month horizon with the hazard layer re-learnt per fold (weights fixed).

Metric	Value
ROC-AUC, deployed rule (Eq. 1–3)	0.673
ROC-AUC, constrained multivariate (5-fold OOF / in-sample)	0.684 / 0.688
ROC-AUC, exploratory unconstrained logistic incl. demographics (5-fold OOF)	0.700
PR-AUC (base rate 0.048)	0.109
ROC-AUC at the 12-month horizon	0.711
Brier, hazard-calibrated $\hat{p}_{12}$ (5-fold OOF)	0.0226
Expected calibration error, $\hat{p}_{12}$ (10 bins over support)	0.003
Precision at recall = 0.80	0.062
Recall in top score decile (lift)	0.288 (2.9×)

escalation) and $\hat{p}_{12}$ (the absolute-risk read, shown alongside). Tier cut-points (20/38/52/66 for Moderate/Elevated/High/Critical) are re-anchored to the deployed score distribution so all five tiers are populated as a monitoring pyramid; operators and jurisdiction rule-packs may override them.

5.4 Performance

Table 2 summarises performance; Figure 3 shows the reliability diagram. Three observations. First, the deployed rule sits within 0.015 AUC of its own multivariate ceiling: the interpretable weighted mean gives up almost nothing relative to the constrained model it is derived from, and the small parameter count (13 bounded weights on $\sim 10^4$ observations) plus the OOF/in-sample agreement (0.684 vs 0.688) indicates negligible overfitting. Second, forgoing demographic features and sign flexibility costs ∼0.027 AUC against the unconstrained reference—the price of monotonicity and demographic neutrality, which we accept by design. Third, these discriminations are consistent with published self-exclusion prediction on comparable features and horizons [7, 8, 10]; under an under-ascertained label [11], measured AUC also *understates* true separation, since an unknown fraction of "negatives" are unlabelled positives [14]. Operationally, a team reviewing the top decile of scores reaches 28.8% of all future RG closures at 2.9× base-rate precision.

6 The staff-observation overlay

Account data cannot see distress, self-disclosed concern, or a family member's phone call—signals that venue and support staff observe directly and that practitioner codes treat as high-priority [12]. The platform therefore maintains a structured interaction log (observations, player-health conversations, interventions, referrals, follow-ups, each tagged with observed markers from a 15-marker taxonomy in five categories: behavioural, financial, time, emotional, social) and folds it into the score as an explicit, bounded overlay—never silently.

Incremental-information weighting. Each observed marker o carries weight ω_o, a kind multiplier $\kappa \in [0.6, 1.0]$ (a referral outranks a passing observation), recency decay $\rho(\Delta t) = 2^{-\Delta t/30\,\text{days}}$,

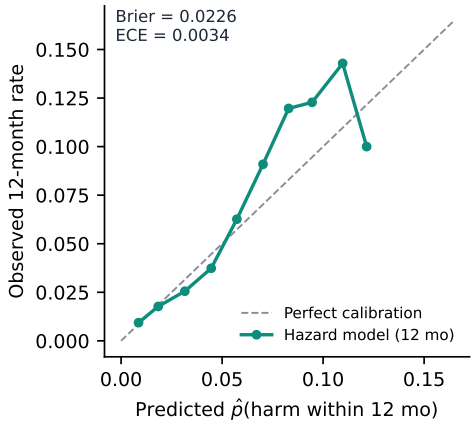


Figure 3: Reliability of the calibrated $\hat{p}$ (5-fold out-of-fold, calibration re-learnt per fold). The in-app model card reports ECE = 0.004 under coarser unit-interval binning; the support-binned value shown here is the stricter figure.

and an *information factor*

$$\iota_o \;=\; \begin{cases} 1 & \text{no data analog (e.g. emotional/social markers)} \\ \max\big(0.15,\; 1 - d_o/100\big) & \text{data analog with sub-score } d_o, \end{cases} \tag{5}$$

so an observation that merely confirms a marker the data already scores highly adds almost nothing, while one contradicting a low data score (a blind spot, e.g. retail cash play) or with no data analog carries full weight. This prevents observation *intensity*—how closely a venue watches—from inflating scores by itself. The total load $L = \sum \omega_o \, \kappa \, \rho \, \iota_o$ saturates into a bounded corroboration uplift $U = 18 \cdot \big(1 - e^{-L/1.9}\big)$, capped so the validated data layer remains primary.

Standing disclosure floor. A critical disclosure (self-disclosed concern; third-party concern) imposes a floor $F = 50$ (the Elevated tier) on the combined score. The floor does not decay with the calendar: it clears only when a staff follow-up records the concern as resolved—and a "checked, no concern" follow-up additionally dampens the corroboration load ($L \leftarrow 0.35L$), making the overlay a two-way ratchet rather than a monotone accumulator of suspicion. The combined score is

$$S^{*} \;=\; \min\big(100,\; \max(S + U,\; F)\big), \tag{6}$$

with S retained, displayed, and separately filterable throughout the product.

The overlay is deliberately *not* statistically fitted—its weights are informed priors—and every governance surface labels it as such (§8). Its effects are individually auditable: each combined score decomposes on screen into "data-derived S + observations U" or "disclosure floor F".

7 Operator deployment

The platform is built so that an RG analyst—not a data team—can go from an operator's raw export to a calibrated deployment.

Cold start. The shipped model (`reference-hazard-2026.07`) works with no operator data at all: weights and calibration come from the reference-cohort fit above, and the demonstration portfolio is synthetic.

Data onboarding. Operators upload up to four CSVs (identities, daily wagers, wallet transactions, sessions). Headers are matched to the canonical schema by an alias dictionary (e.g. `CustomerRef` → `player_id`), common value spellings are coerced (`deposit` → `DEP`; `settled` → `S`; `Male` → `M`), and the analyst confirms the pre-filled mapping with live sample previews—no spreadsheet surgery. Row-level validation and cross-file referential checks run before scoring. An optional `self_excluded` column carries outcome labels.

In-browser recalibration. With labelled data present, the platform re-fits *both* layers—non-negative marker weights (projected gradient descent on Eq. 2) and the calibration—entirely client-side, evaluates on a deterministic 75/25 holdout (ROC-AUC, PR-AUC, Brier, ECE, reliability curve) against the active model, and lets the operator adopt or roll back the result. Operator re-fits currently learn a binary logistic calibration; the time-aware hazard layer (Eq. 4) re-fits once date-labelled outcomes are supplied. When the labelled set is large enough (≥ 40 positives), the re-fit additionally learns *monotone per-marker shapes* — quantile-binned outcome rates smoothed by pool-adjacent-violators — trains a shaped candidate alongside the plain one, and adopts shapes only if they beat the plain fit on the same holdout by at least 0.005 AUC. The guard is not decorative: on the reference cohort an additive-monotone model attains an out-of-fold AUC of 0.679, *no better* than the plain constrained fit (0.684), so there shapes would correctly be rejected; they exist for operator distributions where marker–outcome relationships are materially non-linear. Shapes are always learned from the operator's own distribution — shapes fitted to cohort proxies would not transfer — so the shipped reference model carries none. The adopted model is a portable versioned artifact {weights, calibration, metrics, provenance}; the whole portfolio re-scores on adoption. Because parsing, validation, scoring, and fitting all execute in the browser, **no account-level data leaves the operator's environment**—a deliberately conservative posture for a domain where the training label itself is sensitive.

Benchmark cards. Any deployment can export an anonymised *benchmark card* (model version, weights, shapes, calibration, holdout metrics, cohort size and prevalence—no account-level data), enabling cross-operator comparison of re-fits and an empirical answer to the transportability question of §10 without any data sharing.

8 Governance, auditability, and fairness

Tamper-evident audit. Every score-affecting action (interventions, resolutions, and staff observations) appends to a hash chain,

$$h_k = \text{SHA-256}\big(h_{k-1} \,\|\, \text{canonical}(e_k)\big), \qquad h_0 = 0^{64}, \tag{7}$$

with deterministic canonical serialisation, in the spirit of cryptographic time-stamping [21]. Any retroactive edit or deletion breaks every subsequent link; the product exposes one-click replay verification and exports the chain (with hashes) for regulatory review. The audit log thereby *is* the compliance evidence rather than a mutable table.

Model cards and composition versioning. The governance surface publishes a model card [19]: cohort, design, per-marker weights and evidence, validation metrics, limitations, and version history. Because the shipped score is a composition—validated data layer plus heuristic overlay—the card versions the composition explicitly (`reference-hazard-2026.07 + observed-v1`) and states plainly which layer is validated and which is not. A global toggle switches every surface between the combined score and the validated data-only score.

Fairness. The deployed scorer uses no demographic attributes by construction; fairness is audited post hoc instead: flag-rate parity across subgroups, equal-opportunity recall [20], and per-tier calibration against outcomes. Distinctively, the audit also covers the *observation channel*: because staff attention is discretionary, the platform reports observation rates, mean observed uplift, and disclosure-floor incidence by subgroup—surfacing whether human attention, rather than behaviour, is driving score differences. To our knowledge commercial RG tools do not audit this channel, despite staff identification of problem gamblers being known to be noisy [12].

9 Reference implementation

The reference implementation is a client-side web application (React / TypeScript) structured as ports-and-adapters around a canonical event schema, so ingestion sources, scoring engines, and audit sinks are swappable seams rather than forks. The demonstration portfolio is fully synthetic: 250 deterministic accounts drawn from behavioural archetypes over illustrative, generic aggregate distributions (round values, not derived from the reference cohort or any other dataset), with seeded staff interactions exercising both overlay paths (corroboration and disclosure-driven detection). Under the deployed weights and cut-points the synthetic portfolio occupies all five tiers (79 / 99 / 40 / 31 / 1 from Low to Critical). Beyond the model surfaces, the implementation ships *jurisdiction rule-packs* (per-jurisdiction helpline, minimum age, regulator, and mandatory statement, applied in intervention workflows and the regulator report pack; shipped as illustrative defaults to be verified per deployment) and a *structural game-risk lens* (an instrument scoring games' structural characteristics — event frequency, bet structure, volatility — crossed with player-risk exposure), so game- and player-level risk can be read together. Determinism is enforced throughout (seeded generators; no wall-clock or randomness in scoring paths), so identical inputs always reproduce identical scores, tiers, and audit hashes—a property we consider part of auditability.

10 Limitations and future work

1. **Label validity.** RG-coded closure is an under-ascertained, behaviourally narrow proxy for harm [11]. Weights fitted to it inherit its biases; the PU framing mitigates interpretation but not the label itself. Fitting to richer operator outcomes (complaints, affordability failures, clinical referrals) via the in-browser path is the intended remedy.
2. **Marker sub-scores are heuristics.** Only the weights and calibration are fitted; the $[0, 100]$ marker transforms themselves are rule-based. The re-fit path can now learn monotone per-marker shapes from operator labels (holdout-guarded, §7), but on the reference cohort such shapes add nothing (0.679 vs. 0.684 OOF AUC), so the default transforms remain the rule-based ones.

3. **Three markers are priors.** Session-duration, time-of-day, and in-session loss-of-control markers have no analog in the fitting cohort and carry median weight by assumption. Operators whose data includes sessions can re-fit them out of prior status.

4. **Offline calibration is proxy-based.** The default calibration was fitted on rank-based cohort proxies of the runtime markers; the mapping is exact only after an operator recalibrates on their own scored population, which the product makes a one-click step.

5. **The overlay is unvalidated by design.** Staff-observation weights are informed priors; validating them requires linked observation-outcome data that, to our knowledge, no public dataset provides. The overlay is therefore bounded, labelled, and separable everywhere.

6. **Single-cohort weights.** The default model is fitted on one operator's players from one era of one product mix. Cross-operator transportability is precisely what the re-fit path exists to test; the product exports anonymised benchmark cards (metrics, weights, calibration—no account data) so deployments can compare re-fits as community benchmarks.

11 Conclusion

This reference implementation demonstrates that a responsible-gambling risk engine can be simultaneously *evidence-based* (weights fitted to a longitudinal reference cohort under a leakage-aware design), *interpretable by construction* (a monotone weighted mean whose every contribution is displayable), *calibrated* (an explicit, separately reported probability layer), *honest about its own composition* (validated data layer vs. labelled heuristic overlay, with the human channel audited for bias), and *deployable without data egress*. None of these properties required proprietary methods. We offer the system—and this document—as a reference point for what operators and regulators can reasonably demand of behavioural risk tooling.

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